

	$\mathcal{K}_0^{#1} \dagger$	$\mathcal{K}_0^{#1} \alpha\beta\chi$		
	0	0		
$\mathcal{K}_0^{#1} \dagger^{\alpha\beta\chi}$	0	0	$\mathcal{K}_{1^+}^{#1} \alpha\beta$	$\mathcal{K}_{1^+}^{#2} \alpha\beta$
	$\mathcal{K}_{1^+}^{#1} \dagger^{\alpha\beta}$	$\Upsilon_1 k^2$	$-\frac{\Upsilon_2 k^2}{2\sqrt{2}}$	0
	$\mathcal{K}_{1^+}^{#2} \dagger^{\alpha\beta}$	$-\frac{\Upsilon_2 k^2}{2\sqrt{2}}$	$\Upsilon_3 k^2$	0
	$\mathcal{K}_{1^-}^{#1} \dagger^\alpha$	0	0	$\frac{\Upsilon_4 k^2}{2}$
	$\mathcal{K}_{1^-}^{#2} \dagger^\alpha$	0	0	$-\frac{\Upsilon_5 k^2}{2\sqrt{2}}$
				$\mathcal{K}_{2^+}^{#1} \alpha\beta$
			$\mathcal{K}_{2^+}^{#1} \dagger^{\alpha\beta}$	0
			$\mathcal{K}_{2^-}^{#1} \dagger^{\alpha\beta\chi}$	0
				$\frac{3k^2}{2} \frac{\mathcal{K}_{15}^{(4)}}{\mathcal{K}_{15}}$

	$\mathcal{J}_0^{#1} \dagger$	$\mathcal{J}_0^{#1} \alpha\beta\chi$		
	0	0		
$\mathcal{J}_0^{#1} \dagger^{\alpha\beta\chi}$	0	0	$\mathcal{J}_{1^+}^{#1} \alpha\beta$	$\mathcal{J}_{1^+}^{#2} \alpha\beta$
	$\mathcal{J}_{1^+}^{#1} \dagger^{\alpha\beta}$	$\frac{\Upsilon_3 k^2}{\text{Det}(1^+)}$	$\frac{\Upsilon_2 k^2}{2\sqrt{2}\text{Det}(1^+)}$	0
	$\mathcal{J}_{1^+}^{#2} \dagger^{\alpha\beta}$	$\frac{\Upsilon_2 k^2}{2\sqrt{2}\text{Det}(1^+)}$	$\frac{\Upsilon_1 k^2}{\text{Det}(1^+)}$	0
	$\mathcal{J}_{1^-}^{#1} \dagger^\alpha$	0	0	$\frac{\Upsilon_6 k^2}{2\text{Det}(1^-)}$
	$\mathcal{J}_{1^-}^{#2} \dagger^\alpha$	0	0	$\frac{\Upsilon_5 k^2}{2\sqrt{2}\text{Det}(1^-)}$
				$\mathcal{J}_{2^+}^{#1} \alpha\beta$
			$\mathcal{J}_{2^+}^{#1} \dagger^{\alpha\beta}$	0
			$\mathcal{J}_{2^-}^{#1} \dagger^{\alpha\beta\chi}$	0
				$\frac{2}{3k^2} \frac{\mathcal{K}_{15}^{(4)}}{\mathcal{K}_{15}}$

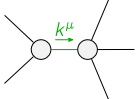
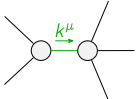
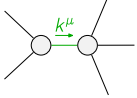
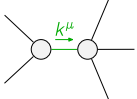
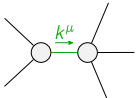
Abbreviations used in matrices

$$\begin{aligned} \Upsilon_1 &== 2 \mathcal{K}_{15}^{(4)} - \mathcal{K}_3^{(4)} \ \&\& \ \Upsilon_2 == 2 \mathcal{K}_{15}^{(4)} - \mathcal{K}_4^{(4)} \ \&\& \ \Upsilon_3 == \mathcal{K}_{15}^{(4)} + \mathcal{K}_9^{(4)} \ \&\& \\ \Upsilon_4 &== 2 \mathcal{K}_1^{(4)} + 3 \mathcal{K}_{15}^{(4)} \ \&\& \ \Upsilon_5 == 2 \mathcal{K}_1^{(4)} - \mathcal{K}_7^{(4)} \ \&\& \ \Upsilon_6 == \mathcal{K}_1^{(4)} + 3 \mathcal{K}_{15}^{(4)} - \mathcal{K}_3^{(4)} - \mathcal{K}_4^{(4)} - \mathcal{K}_7^{(4)} + 2 \mathcal{K}_9^{(4)} \ \&\& \\ \text{Det}(1^+) &== \frac{1}{8} k^4 (12 \mathcal{K}_{15}^{(4)2} - \mathcal{K}_4^{(4)2} - 8 \mathcal{K}_3^{(4)} \mathcal{K}_9^{(4)} + 4 \mathcal{K}_{15}^{(4)} (-2 \mathcal{K}_3^{(4)} + \mathcal{K}_4^{(4)} + 4 \mathcal{K}_9^{(4)})) \end{aligned}$$

Lagrangian

$$\begin{aligned} &\mathcal{K}_1^{(4)} \partial_\beta \mathcal{K}_{\chi\delta}^\delta \partial^\chi \mathcal{K}_\alpha^{\alpha\beta} - \mathcal{K}_1^{(4)} \partial_\chi \mathcal{K}_{\beta\delta}^\delta \partial^\chi \mathcal{K}_\alpha^{\alpha\beta} + \mathcal{K}_3^{(4)} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\alpha\chi}^\delta + \\ &\mathcal{K}_4^{(4)} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\beta\chi}^\delta + 3 \mathcal{K}_{15}^{(4)} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\chi\alpha}^\delta - \mathcal{K}_3^{(4)} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\chi\alpha}^\delta + \\ &\mathcal{K}_7^{(4)} \partial^\chi \mathcal{K}_\alpha^{\alpha\beta} \partial_\delta \mathcal{K}_{\chi\beta}^\delta + \mathcal{K}_9^{(4)} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\beta\chi}^\delta + \mathcal{K}_{15}^{(4)} \partial_\delta \mathcal{K}_{\alpha\beta\chi} \partial^\delta \mathcal{K}^{\alpha\beta\chi} + \mathcal{K}_{15}^{(4)} \partial_\delta \mathcal{K}_{\beta\alpha\chi} \partial^\delta \mathcal{K}^{\alpha\beta\chi} \end{aligned}$$

Added source term(s):	$\mathcal{K}^{\alpha\beta\chi} \mathcal{J}_{\alpha\beta\chi}$	
Source constraint(s)	# constraint(s)	Covariant form
$\mathcal{J}_0^{#1\alpha\beta\chi} == 0$	1	$\partial_\delta \partial^\alpha \mathcal{J}^{\beta\chi\delta} + \partial_\delta \partial^\alpha \mathcal{J}^{\delta\beta\chi} + \partial_\delta \partial^\beta \mathcal{J}^{\chi\alpha\delta} + \partial_\delta \partial^\chi \mathcal{J}^{\alpha\beta\delta} + \partial_\delta \partial^\chi \mathcal{J}^{\delta\alpha\beta} + \partial_\delta \partial^\delta \mathcal{J}^{\beta\alpha\chi} ==$ $\partial_\delta \partial^\alpha \mathcal{J}^{\chi\beta\delta} + \partial_\delta \partial^\beta \mathcal{J}^{\alpha\chi\delta} + \partial_\delta \partial^\beta \mathcal{J}^{\delta\alpha\chi} + \partial_\delta \partial^\chi \mathcal{J}^{\beta\alpha\delta} + \partial_\delta \partial^\delta \mathcal{J}^{\alpha\beta\chi} + \partial_\delta \partial^\delta \mathcal{J}^{\chi\alpha\beta}$
$\mathcal{J}_0^{#1} == 0$	1	$\partial_\beta \mathcal{J}_\alpha^{\alpha\beta} == 0$
$\mathcal{J}_{2^+}^{#1\alpha\beta} == 0$	5	$3 \partial_\delta \partial_\chi \partial^\alpha \mathcal{J}^{\chi\beta\delta} + 3 \partial_\delta \partial_\chi \partial^\beta \mathcal{J}^{\chi\alpha\delta} + 2 \eta^{\alpha\beta} \partial_\epsilon \partial^\epsilon \partial_\delta \mathcal{J}_\chi^{\chi\delta} == 2 \partial_\delta \partial^\beta \partial^\alpha \mathcal{J}_\chi^{\chi\delta} + 3 (\partial_\delta \partial^\delta \partial_\chi \mathcal{J}^{\alpha\beta\chi} + \partial_\delta \partial^\delta \partial_\chi \mathcal{J}^{\beta\alpha\chi})$
Total # constraint(s):	7	

Resolved pole(s)	# polarization(s)	Square mass	Residue
	1	0	$\begin{aligned} &(432 \mathcal{K}_{15}^{(4)4} + \mathcal{K}_4^{(4)2} \mathcal{K}_7^{(4)2} - 36 \mathcal{K}_{15}^{(4)3} (12 \mathcal{K}_3^{(4)} + \mathcal{K}_4^{(4)} + 4 (\mathcal{K}_7^{(4)} - 4 \mathcal{K}_9^{(4)})) + 8 \mathcal{K}_3^{(4)} \mathcal{K}_7^{(4)2} \mathcal{K}_9^{(4)} + \\ &6 \mathcal{K}_{15}^{(4)2} (16 \mathcal{K}_3^{(4)2} - 9 \mathcal{K}_4^{(4)2} + 2 \mathcal{K}_3^{(4)} (5 \mathcal{K}_4^{(4)} + 8 \mathcal{K}_7^{(4)} - 36 \mathcal{K}_9^{(4)}) - 2 \mathcal{K}_4^{(4)} (3 \mathcal{K}_7^{(4)} + 2 \mathcal{K}_9^{(4)}) - 4 (\mathcal{K}_7^{(4)2} + 4 \mathcal{K}_7^{(4)} \mathcal{K}_9^{(4)} - 8 \mathcal{K}_9^{(4)2})) - \\ &\sqrt{((-18 \mathcal{K}_1^{(4)} \mathcal{K}_{15}^{(4)} - 18 \mathcal{K}_{15}^{(4)2} + \mathcal{K}_7^{(4)2} + 4 \mathcal{K}_1^{(4)} (\mathcal{K}_3^{(4)} + \mathcal{K}_4^{(4)} - 2 \mathcal{K}_9^{(4)}) + 6 \mathcal{K}_{15}^{(4)} (\mathcal{K}_3^{(4)} + \mathcal{K}_4^{(4)} + \mathcal{K}_7^{(4)} - 2 \mathcal{K}_9^{(4)}))^2} \\ &(864 \mathcal{K}_{15}^{(4)4} - 384 \mathcal{K}_{15}^{(4)3} (3 \mathcal{K}_3^{(4)} - \mathcal{K}_4^{(4)} - 2 \mathcal{K}_9^{(4)}) + 3 (\mathcal{K}_4^{(4)2} + 8 \mathcal{K}_3^{(4)} \mathcal{K}_9^{(4)})^2 + \\ &8 \mathcal{K}_{15}^{(4)2} (48 \mathcal{K}_3^{(4)2} + \mathcal{K}_4^{(4)2} + 42 \mathcal{K}_4^{(4)} \mathcal{K}_9^{(4)} + 98 \mathcal{K}_9^{(4)2} - 16 \mathcal{K}_3^{(4)} (2 \mathcal{K}_4^{(4)} + 7 \mathcal{K}_9^{(4)})) + \\ &4 \mathcal{K}_{15}^{(4)} (64 \mathcal{K}_3^{(4)2} \mathcal{K}_9^{(4)} - \mathcal{K}_4^{(4)2} (5 \mathcal{K}_4^{(4)} + 24 \mathcal{K}_9^{(4)}) + 8 \mathcal{K}_3^{(4)} (\mathcal{K}_4^{(4)2} - 5 \mathcal{K}_4^{(4)} \mathcal{K}_9^{(4)} - 24 \mathcal{K}_9^{(4)2})))) + \\ &2 \mathcal{K}_1^{(4)} (216 \mathcal{K}_{15}^{(4)3} + 6 \mathcal{K}_{15}^{(4)2} (-32 \mathcal{K}_3^{(4)} + \mathcal{K}_4^{(4)} + 40 \mathcal{K}_9^{(4)}) + \mathcal{K}_{15}^{(4)} (32 \mathcal{K}_3^{(4)2} - 21 \mathcal{K}_4^{(4)2} + 4 \mathcal{K}_3^{(4)} (5 \mathcal{K}_4^{(4)} - 42 \mathcal{K}_9^{(4)}) - \\ &8 \mathcal{K}_4^{(4)} \mathcal{K}_9^{(4)} + 64 \mathcal{K}_9^{(4)2}) + 2 (\mathcal{K}_4^{(4)2} (\mathcal{K}_4^{(4)} - 2 \mathcal{K}_9^{(4)}) + 8 \mathcal{K}_3^{(4)2} \mathcal{K}_9^{(4)} + \mathcal{K}_3^{(4)} (\mathcal{K}_4^{(4)2} + 8 \mathcal{K}_4^{(4)} \mathcal{K}_9^{(4)} - 16 \mathcal{K}_9^{(4)2}))) + \\ &2 \mathcal{K}_{15}^{(4)} (3 \mathcal{K}_4^{(4)3} - 3 \mathcal{K}_4^{(4)} \mathcal{K}_7^{(4)2} + 3 \mathcal{K}_4^{(4)2} (\mathcal{K}_7^{(4)} - 2 \mathcal{K}_9^{(4)}) + 24 \mathcal{K}_3^{(4)2} \mathcal{K}_9^{(4)} - 8 \mathcal{K}_7^{(4)2} \mathcal{K}_9^{(4)} + \\ &\mathcal{K}_3^{(4)} (3 \mathcal{K}_4^{(4)2} + 24 \mathcal{K}_4^{(4)} \mathcal{K}_9^{(4)} + 8 (\mathcal{K}_7^{(4)2} + 3 \mathcal{K}_7^{(4)} \mathcal{K}_9^{(4)} - 6 \mathcal{K}_9^{(4)2})))))/ \\ &(\mathcal{K}_{15}^{(4)} (18 \mathcal{K}_1^{(4)} \mathcal{K}_{15}^{(4)} + 18 \mathcal{K}_{15}^{(4)2} - \mathcal{K}_7^{(4)2} - 4 \mathcal{K}_1^{(4)} (\mathcal{K}_3^{(4)} + \mathcal{K}_4^{(4)} - 2 \mathcal{K}_9^{(4)}) - 6 \mathcal{K}_{15}^{(4)} (\mathcal{K}_3^{(4)} + \mathcal{K}_4^{(4)} + \mathcal{K}_7^{(4)} - 2 \mathcal{K}_9^{(4)})) \\ &(12 \mathcal{K}_{15}^{(4)2} - \mathcal{K}_4^{(4)2} - 8 \mathcal{K}_3^{(4)} \mathcal{K}_9^{(4)} + 4 \mathcal{K}_{15}^{(4)} (-2 \mathcal{K}_3^{(4)} + \mathcal{K}_4^{(4)} + 4 \mathcal{K}_9^{(4)}))) \end{aligned}$
	1	0	$\begin{aligned} &(432 \mathcal{K}_{15}^{(4)4} + \mathcal{K}_4^{(4)2} \mathcal{K}_7^{(4)2} - 36 \mathcal{K}_{15}^{(4)3} (12 \mathcal{K}_3^{(4)} + \mathcal{K}_4^{(4)} + 4 (\mathcal{K}_7^{(4)} - 4 \mathcal{K}_9^{(4)})) + 8 \mathcal{K}_3^{(4)} \mathcal{K}_7^{(4)2} \mathcal{K}_9^{(4)} + \\ &6 \mathcal{K}_{15}^{(4)2} (16 \mathcal{K}_3^{(4)2} - 9 \mathcal{K}_4^{(4)2} + 2 \mathcal{K}_3^{(4)} (5 \mathcal{K}_4^{(4)} + 8 \mathcal{K}_7^{(4)} - 36 \mathcal{K}_9^{(4)}) - 2 \mathcal{K}_4^{(4)} (3 \mathcal{K}_7^{(4)} + 2 \mathcal{K}_9^{(4)}) - 4 (\mathcal{K}_7^{(4)2} + 4 \mathcal{K}_7^{(4)} \mathcal{K}_9^{(4)} - 8 \mathcal{K}_9^{(4)2})) + \\ &\sqrt{((-18 \mathcal{K}_1^{(4)} \mathcal{K}_{15}^{(4)} - 18 \mathcal{K}_{15}^{(4)2} + \mathcal{K}_7^{(4)2} + 4 \mathcal{K}_1^{(4)} (\mathcal{K}_3^{(4)} + \mathcal{K}_4^{(4)} - 2 \mathcal{K}_9^{(4)}) + 6 \mathcal{K}_{15}^{(4)} (\mathcal{K}_3^{(4)} + \mathcal{K}_4^{(4)} + \mathcal{K}_7^{(4)} - 2 \mathcal{K}_9^{(4)}))^2} \\ &(864 \mathcal{K}_{15}^{(4)4} - 384 \mathcal{K}_{15}^{(4)3} (3 \mathcal{K}_3^{(4)} - \mathcal{K}_4^{(4)} - 2 \mathcal{K}_9^{(4)}) + 3 (\mathcal{K}_4^{(4)2} + 8 \mathcal{K}_3^{(4)} \mathcal{K}_9^{(4)})^2 + \\ &8 \mathcal{K}_{15}^{(4)2} (48 \mathcal{K}_3^{(4)2} + \mathcal{K}_4^{(4)2} + 42 \mathcal{K}_4^{(4)} \mathcal{K}_9^{(4)} + 98 \mathcal{K}_9^{(4)2} - 16 \mathcal{K}_3^{(4)} (2 \mathcal{K}_4^{(4)} + 7 \mathcal{K}_9^{(4)})) + \\ &4 \mathcal{K}_{15}^{(4)} (64 \mathcal{K}_3^{(4)2} \mathcal{K}_9^{(4)} - \mathcal{K}_4^{(4)2} (5 \mathcal{K}_4^{(4)} + 24 \mathcal{K}_9^{(4)}) + 8 \mathcal{K}_3^{(4)} (\mathcal{K}_4^{(4)2} - 5 \mathcal{K}_4^{(4)} \mathcal{K}_9^{(4)} - 24 \mathcal{K}_9^{(4)2})))) + \\ &2 \mathcal{K}_1^{(4)} (216 \mathcal{K}_{15}^{(4)3} + 6 \mathcal{K}_{15}^{(4)2} (-32 \mathcal{K}_3^{(4)} + \mathcal{K}_4^{(4)} + 40 \mathcal{K}_9^{(4)}) + \mathcal{K}_{15}^{(4)} (32 \mathcal{K}_3^{(4)2} - 21 \mathcal{K}_4^{(4)2} + 4 \mathcal{K}_3^{(4)} (5 \mathcal{K}_4^{(4)} - 42 \mathcal{K}_9^{(4)}) - \\ &8 \mathcal{K}_4^{(4)} \mathcal{K}_9^{(4)} + 64 \mathcal{K}_9^{(4)2}) + 2 (\mathcal{K}_4^{(4)2} (\mathcal{K}_4^{(4)} - 2 \mathcal{K}_9^{(4)}) + 8 \mathcal{K}_3^{(4)2} \mathcal{K}_9^{(4)} + \mathcal{K}_3^{(4)} (\mathcal{K}_4^{(4)2} + 8 \mathcal{K}_4^{(4)} \mathcal{K}_9^{(4)} - 16 \mathcal{K}_9^{(4)2}))) + \\ &2 \mathcal{K}_{15}^{(4)} (3 \mathcal{K}_4^{(4)3} - 3 \mathcal{K}_4^{(4)} \mathcal{K}_7^{(4)2} + 3 \mathcal{K}_4^{(4)2} (\mathcal{K}_7^{(4)} - 2 \mathcal{K}_9^{(4)}) + 24 \mathcal{K}_3^{(4)2} \mathcal{K}_9^{(4)} - 8 \mathcal{K}_7^{(4)2} \mathcal{K}_9^{(4)} + \\ &\mathcal{K}_3^{(4)} (3 \mathcal{K}_4^{(4)2} + 24 \mathcal{K}_4^{(4)} \mathcal{K}_9^{(4)} + 8 (\mathcal{K}_7^{(4)2} + 3 \mathcal{K}_7^{(4)} \mathcal{K}_9^{(4)} - 6 \mathcal{K}_9^{(4)2})))))/ \\ &(\mathcal{K}_{15}^{(4)} (18 \mathcal{K}_1^{(4)} \mathcal{K}_{15}^{(4)} + 18 \mathcal{K}_{15}^{(4)2} - \mathcal{K}_7^{(4)2} - 4 \mathcal{K}_1^{(4)} (\mathcal{K}_3^{(4)} + \mathcal{K}_4^{(4)} - 2 \mathcal{K}_9^{(4)}) - 6 \mathcal{K}_{15}^{(4)} (\mathcal{K}_3^{(4)} + \mathcal{K}_4^{(4)} + \mathcal{K}_7^{(4)} - 2 \mathcal{K}_9^{(4)})) \\ &(12 \mathcal{K}_{15}^{(4)2} - \mathcal{K}_4^{(4)2} - 8 \mathcal{K}_3^{(4)} \mathcal{K}_9^{(4)} + 4 \mathcal{K}_{15}^{(4)} (-2 \mathcal{K}_3^{(4)} + \mathcal{K}_4^{(4)} + 4 \mathcal{K}_9^{(4)}))) \end{aligned}$
	2	0	(Hidden for brevity)
	2	0	(Hidden for brevity)
	2	0	(Hidden for brevity)